Enhancing Data Security in GA4GH Task Execution Services with Confidential Computing

Pavel Nikonorov[1], Ruslan Vakhitov[1], Alexander Senf[2], Boris Guennewig[2], Alex Kanitz[3]

[1] GENXT, Wellcome Genome Campus, Hinxton, UK; [2] Pacific Analytics, Sydney, Australia; [3] Swiss Institute of Bioinformatics / ELIXIR Switzerland

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Pacific Analytics GENXT

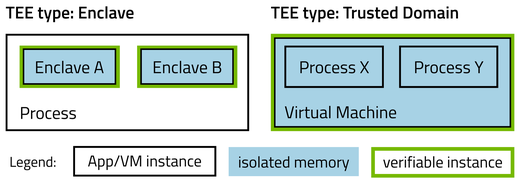
www.pacificanalytics.com.au **www.genxt.network**

CONFIDENTIAL COMPUTING OVERVIEW

In conventional computing systems, sharing of data with another party implies its disclosure and the inability to control its further usage. Confidential Computing is a CPU-based technology that addresses these issues by enabling hardware-enforced trust establishment between remote computing devices and allowing them to exchange data that cannot be accessed outside one's environment. The technology provides a new security feature for the protection of data in use by performing computations in a hardware-based, isolated and attested Trusted Execution Environment (TEE). Pioneered by Intel, Confidential Computing marks an evolutionary leap, transitioning from TEE in mobile devices to high-performance general-purpose CPUs, SoC chips, and now to GPUs. intel. AMD arm NVIDIA.

TRUSTED EXECUTION ENVIRONMENT / TEE

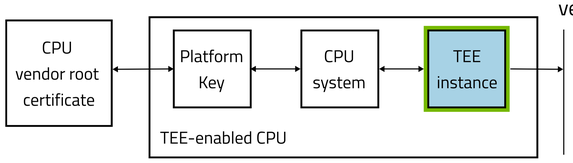
TEE provides an isolated execution environment for user-defined code and data, i.e., the confidentiality and integrity of code and data processed in this environment are protected against other software and physical access. TEE implementations can be categorised as Enclaves and Trusted Domains. An enclave is a protected area within the process (Intel SGX). One can use open-source TEE SDKs to leverage TEE functions or use auxiliary software (so-called LibOS) to execute existing legacy applications inside enclaves. For example, some frameworks provide pathed Python or Go interpreters with a transparent TEE support. In contrast, Trusted Domains offers fully-encrypted isolated and attestable Virtual Machines (Intel TDX, AMD SEV/SEV-SNP, ARM TrustZone). With both types, TEE protects process/VM memory from direct external access, so developers should still take care of inputs and outputs defined by the software.



TEE-based software instances may work with network socket connections, file system objects, and are able to *seal data*, ensuring that encrypted information is exclusively accessible by the permitted TEEs.

TEE REMOTE ATTESTATION

Remote attestation is a method by which a host authenticates it's hardware and software configuration to a remote host. The goal of remote attestation is to enable a remote system to validate the integrity and trustworthiness of another system. RA involves generating verifiable cryptographic evidence or measurement (the so-called Attestation Quote or Report) by requesting relevant CPU instructions from inside the TEE. The evidence is signed by the CPU's Platform Key (PK) which is fused during the manufacturing process and signed by the CPU vendor. PK is only accessible by the CPU.



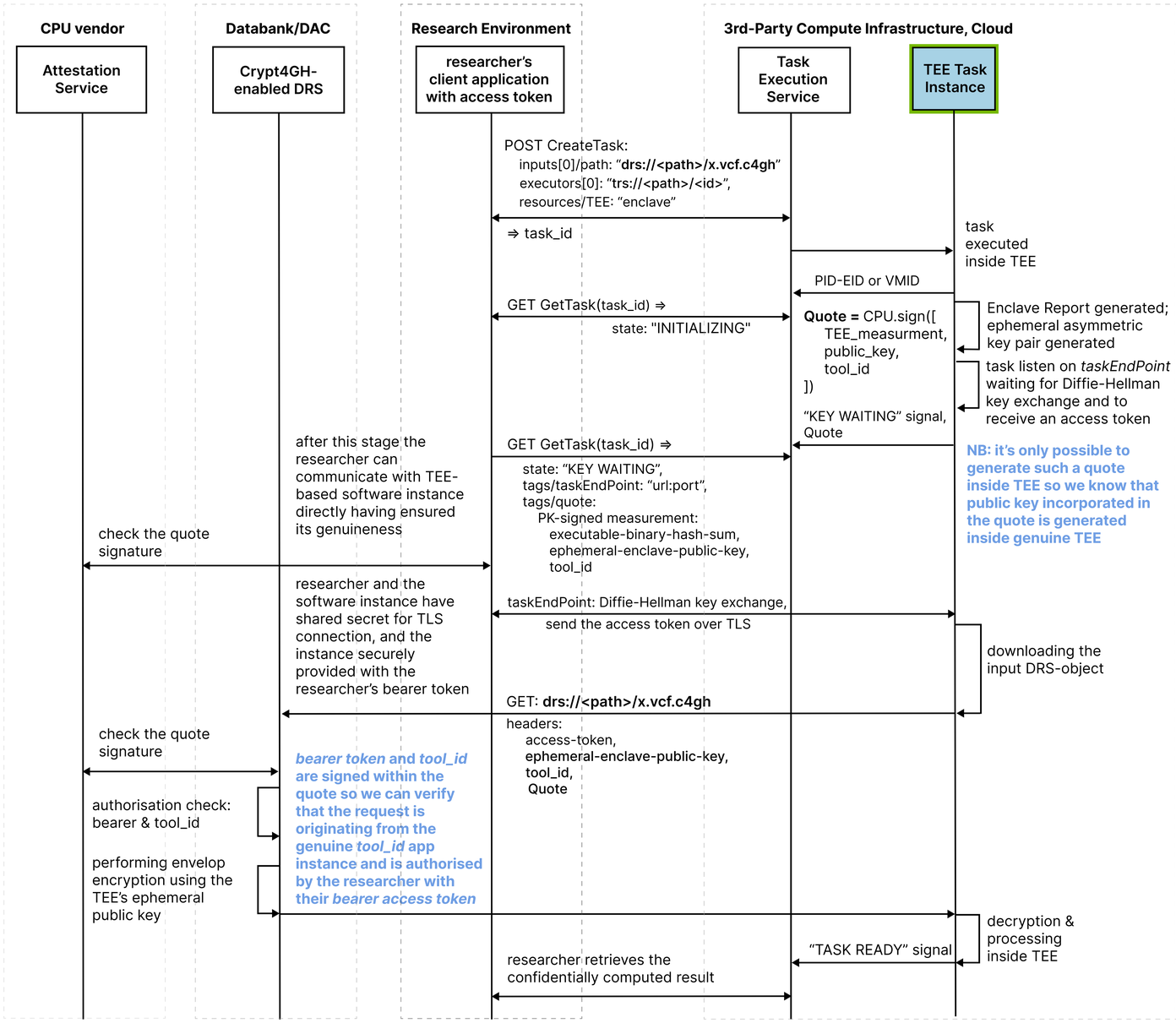
verifiable cryptographic evidence that contain:

- CPU platform version
- code and data measurement
- developer signature (optional)
- ephemeral public key (optional)

Software instance can send evidence to a remote party, which can validate its signature to conclude if the remote code is running on a genuine CPU inside the TEE. The measurement also contains a digest of the code and static data inside this TEE which allows the relying party to confirm that the remote code has not been tampered with. Evidence may also contain the developer signature, and the public key of a key pair generated inside the TEE, which allows the remote party to provision secrets confidentially to the TEE without the need to expose data and to trust any other software running on the server.

TEE-BASED TES WITH CRYPT4GH-ENABLED DRS AND TRS

Confidential Computing offers a promising approach to enhance the GA4GH API implementations by introducing a comprehensive protection of both integrity and confidentiality in federated environments. We propose a specification draft of the TEE-enabled GA4GH TES and Crypt4GH-enabled DRS that provides two advantages. Firstly, it offers an ability to perform computations in complete privacy, even in public infrastructures. Secondly, when integrated with the TRS, and GA4GH Machine-Readable Consents, it can enforce data owner consent policies by ensuring that researchers are allowed to use approved tools only to process permitted datasets inside the TEE without disclosure.



DISCUSSION

The TEE-based TES proposes hardware-enforced, fine-grained data security for federated analysis without data disclosure, reflecting the dual needs of privacy and policy compliance, which can lay the groundwork for the next generation of Trusted Research Environments.